

Dual role of the IL-12/IFN-gamma axis in the development of autoimmune myocarditis: induction by IL-12 and protection by IFN-gamma.

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IL-12 and IFN-gamma positively regulate each other and type 1 inflammatory responses, which are believed to cause tissue damage in autoimmune diseases. We investigated the role of the IL-12/IFN-gamma (Th1) axis in the development of autoimmune myocarditis. IL-12p40-deficient mice on a susceptible background resisted myocarditis. In the absence of IL-12, autospecific CD4(+) T cells proliferated poorly and showed increased Th2 cytokine responses. However, IFN-gamma-deficient mice developed fatal autoimmune disease, and blockade of IL-4R signaling did not confer susceptibility to myocarditis in IL-12p40-deficient mice, demonstrating that IL-12 triggers autoimmunity by a mechanism independent of the effector cytokines IFN-gamma and IL-4. In conclusion, our results suggest that the IL-12/IFN-gamma axis is a double-edged sword for the development of autoimmune myocarditis. Although IL-12 mediates disease by induction/expansion of Th1-type cells, IFN-gamma production from these cells limits disease progression.